



Jewelry That Cares: Harnessing Smart Jewelry for Emotional Well-being

Capstone Project Phase A

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Link to Github:

<https://github.com/Alaa-Barazi/Smart-Jewelry>



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Abstract

Anxiety disorders significantly impact daily functioning and quality of life, yet existing therapeutic solutions cannot provide real-time support during daily life. Current market applications focus on either psychological support or physiological tracking, but rarely integrate both with professional therapist involvement. This project develops "Smart Jewelry" , a comprehensive anxiety-support system that combines automatic anxiety detection through wearable sensors, immediate haptic-based intervention, and integrated therapist oversight with community support. Through expert interviews with eight mental health professionals, we established clinical requirements for the system. Phase A delivers competitive analysis, system architecture design, central functional and non-functional requirements, and a technology stack leveraging React Native, Firebase, and Bluetooth communication. Phase B will implement and clinically validate the system to demonstrate reduced anxiety episodes and improved patient outcomes.

1.Introduction

Anxiety disorders are among the most prevalent mental health challenges globally, significantly impacting individuals' daily functioning, well-being, and overall quality of life. As this societal burden grows, the need for continuous, accessible support beyond traditional clinical settings becomes increasingly urgent (Bhimani et al., 2025). While conventional therapies are highly effective, they often cannot provide the immediate, in-the-moment assistance needed during daily life. Recent advancements in wearable technology and artificial intelligence offer a promising solution, driving a shift from passive health monitoring to smart, responsive systems that actively assist users in real-time.

The foundation of an effective mental health support system relies on the ability to continuously and objectively detect anxiety. By utilizing modern smart devices equipped with physiological sensors - such as heart rate and electrodermal activity monitors - combined with advanced machine learning algorithms, it is now possible to accurately identify stress and anxiety even in noisy, real-world environments. This allows for personalized, continuous tracking outside of controlled clinical conditions.

However, accurate detection is only the first step. To provide meaningful support, technology must move from merely alerting the user to actively intervening. Studies demonstrate that haptic technology, specifically vibration-based feedback, is a highly effective and user-friendly method for reducing anxiety (Fason et al., 2025; Terenti & Vatavu, 2025). By combining real-time physiological monitoring with optimal wearable design, it is possible to create an emotionally intelligent, closed-loop system (Mishra et al., 2025). Such a system can automatically trigger calming physical sensations the exact moment an anxiety spike is detected, helping the user regulate their emotions instantly.

Furthermore, for these technological solutions to reach their full potential, they must be part of a holistic care plan. Integrating the objective physiological data collected by

wearables with subjective user reporting through mobile applications creates a comprehensive tool (Wang et al., 2026). This integration bridges the gap between daily technological intervention and traditional psychological treatments, such as group Cognitive Behavioral Therapy (CBT), providing a complete support ecosystem.

Based on these insights, this project develops a comprehensive anxiety-support system that combines real-world AI detection, immediate haptic intervention, and therapeutic app integration to provide continuous, personalized care. To realize this vision, we introduce a mobile application featuring secure documentation, gamification, therapist metrics, and improved patient-therapist connectivity. The remainder of this book outlines the platform's development, examining its technological, organizational (including UML and architectural design), and ethical aspects.

2. Background

Existing solutions:

We performed a search in app stores and published literature and found several leading applications for anxiety disorders:

Rootd(<https://www.rootd.io/>):

An app that has a button that you press when the patient is having an anxiety attack. It offers them breathing exercises and cognitive methods. In addition, there is a kind of training magazine on anxiety attacks, teaching the patient what they can do in the short term to improve as well as in the long term.

Patient side only, no therapist side.

Feeltherapeutics(<https://www.feeltherapeutics.com/>):

An app that is connected to a bracelet that measures temperature, heartbeat, and more. The app detects anxiety attacks, depression. Saves moods every day, breathing exercises, allows the patient to record notes/thoughts.

Sanvello(<https://pacifica.en.aptoide.com/app>):

An app that integrates the community to support people going through an anxiety attack. The patient can write their feelings in the app and anyone who sees can respond and support. In addition, there is a therapist side that you can talk to all the time. For an additional fee, you can receive 1:1 coaching with a personal therapist to set and achieve goals. Based primarily on CBT treatments.

Limited to the US only.

Table 1. Comparison of Existing Anxiety Management Solutions

To contextualize our project within the existing market, we compared three leading anxiety management applications across key functional features. The following table illustrates how each solution approaches patient support, therapist integration, and monitoring capabilities.

	Rootd	feeltherapeutics	Sanvello	Our application
therapist (professional) side	✗	✗	✓	✓
auto anxiety detection	✗	✓	✗	✓
Documentation experience	✗	✓	✓	✓
Chat with AI	✗	✗	✗	✓
Chat/Call therapist	✗	✓	✓	✓
Connected with fidget	✗	✓	✗	✓
Future improvement (Extra information)	✓	✗	✓	✓
Therapy with music / sounds	✗	✗	✗	✓
Community Support	✗	✗	✓	✓

Gap in Existing Solutions:

As demonstrated in the comparison above, current market applications tend to focus on either psychological support or physiological tracking, but rarely both. For instance, while Sanvello offers strong therapeutic features and community support, it completely lacks automatic anxiety detection. Conversely, feeltherapeutics provides excellent

wearable-based detection but does not integrate professional therapist support or community engagement. Therefore, there is a distinct gap in the market for a holistic, unified platform

3.Literature review

3.1 AI and Wearables for Anxiety Detection

Several studies lay the theoretical and technological groundwork for detecting anxiety using physiological markers. Castillo Zorro et al. (2026) provide a comprehensive foundation through a systematic review of 77 studies, examining how electronic devices and mobile applications utilize physiological and behavioral data to detect anxiety, emphasizing the importance of specific sensors, device placement, and the need for personalization. Jolly et al. (2025) strengthens this foundation by reviewing the specific modern smart gadgets capable of tracking these markers, demonstrating how real-time physiological signals like heart rate, respiration, and electrodermal activity (EDA) are collected. Building upon this theoretical and hardware basis, Alkurdi et al. (2025) takes the research a step further into practical application. They investigate anxiety detection using multimodal wearable data and demonstrate that applying machine learning models to these signals (such as ECG and EDA) allows for reliable performance even in real-world, noisy conditions. Finally, Bhimani et al. (2025) expands on this capability by presenting an AI-driven, wearable-based system that continuously and objectively monitors changes in mental conditions outside of clinical settings, proving that automatic real-time detection in everyday life is highly feasible.

3.2 Haptic Feedback for Anxiety Intervention

Moving beyond the challenge of detection, researchers have explored how wearable technology can actively intervene to soothe users, particularly through haptic (vibration) feedback. Fason et al. (2025) lays the clinical groundwork for this approach, evaluating a wearable haptic patch and demonstrating that vibration-based stimulation significantly reduces anxiety levels while achieving high user satisfaction. To effectively implement such a solution, Terenti & Vatavu (2025) adds a crucial usability layer; their study explores interaction techniques combining touchscreens with wearable haptic feedback, providing essential insights into optimal device placement and vibrotactile patterns to enhance the calming user experience. Bringing the concepts of detection and intervention together, Mishra et al. (2025) builds upon the proven clinical effectiveness of haptics and optimal UX design to introduce an "emotionally intelligent haptic system." Their closed-loop system automatically detects anxiety in real-time (using EEG and machine learning) and immediately delivers haptic feedback, allowing the user to actively regulate their anxiety exactly when it occurs.

3.3 Integrating Wearables & Apps in Therapy

To understand how these technological tools fit into a comprehensive care plan, Wang et al. (2026) places these advancements into a broader therapeutic context. While the previous studies focus on hardware, AI, and immediate intervention, Wang et al. investigates the holistic approach. Their study demonstrates that objective data collected by wearables (such as heart rate and activity levels), when combined with subjective daily mood and energy reports via a mobile app, can successfully support traditional, long-term psychological treatments like group Cognitive Behavioral Therapy (CBT) (Wang et al., 2026).

Together, these studies establish that real-time, accurate anxiety detection using wearable sensors and machine learning is technically feasible and clinically validated. However, detection alone is insufficient for therapeutic impact users also require immediate, effective intervention strategies to manage anxiety in real-time, which motivates the need for integrated haptic feedback systems explored in the following section.

4. Engineering process

Our project includes the following working stages:

Phase A: Research & Planning establishes the foundation by conducting market research, needs assessment through expert interviews, and architectural planning to identify the gap in existing anxiety-management solutions.

Phase B: Development & Implementation moves into practical realization, focusing on hardware-software integration, testing success measures, and clinical evaluation to validate the system's effectiveness in reducing anxiety frequency and improving user outcomes.

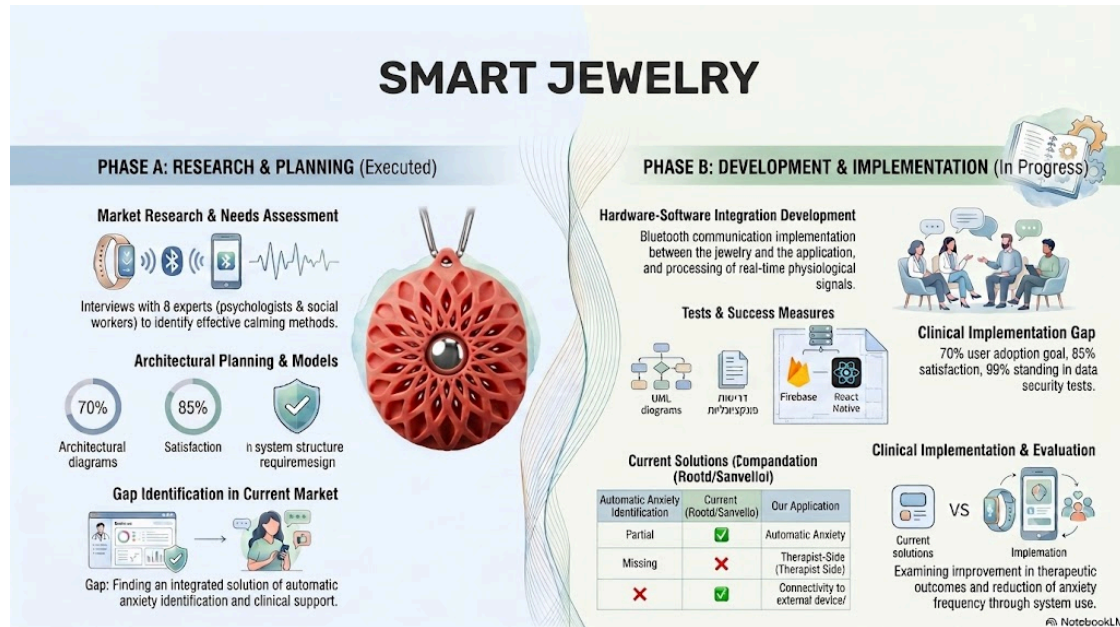


fig 1. Project workflow

Figure 1 represents a visual summary of the project's two-phase process, contrasting the research and requirements work of Phase A with the planned technical implementation of Phase B, including the system architecture, hardware integration, and success metrics. It also highlights how the proposed application compares to existing market solutions such as Rootd and Sanvello.

4.1 Interviews

Developing a sensitive and effective anxiety-management system requires grounding the technology in clinical practice. By consulting mental health professionals, we gained crucial insights into patient behaviors and treatment challenges to define the system's functionality and user experience. To develop responsible AI requirements for a smart fidget system that detects anxiety-related fidgeting patterns, we analyzed both human-based and AI-based requirements elicitation processes. The study included interviews with eight mental health experts: four clinical psychologists (P1–P4) working in private and hospital settings, two hospital social workers (SW1–SW2) in Western populations, and two welfare-sector social workers (W1–W2) in non-Western populations.

Two AI informants were included to address the single-interview limitation and to test whether bias profiles replicate across models and demographic configurations. AI1 was a persona generated by Gemini, presented as a male, 48-year-old clinical psychologist operating a private clinic, specializing in CBT , biofeedback, and mindfulness. AI2 was a persona generated by Claude, presented as a female, 41-year-old clinical social worker in a public hospital setting, specializing in DBT and psychosocial approaches . The AI2 persona was designed to differ from AI1 on every demographic axis that AI1's profile was found to skew gender, sector, therapeutic modality, and institutional context. This contrastive design tests whether the bias patterns identified in AI1 are artefacts of a specific prompt or structural properties of LLM-generated informants. We acknowledge that using Claude to generate the AI2 transcript could itself be seen as an instance of Lin's substitution fallacy. We treat this as a methodological constraint rather than a confound: the AI2 interview was used not only as a source of clinical knowledge, but mainly as a controlled stimulus for bias auditing, and the finding that AI2 reproduced AI1's structural absences despite different prompts is analytically informative regardless of which model generated it. Table 2 presents interviewers's ' details:

Table 2: Interviewees' details

Code	Role	Gender	Sector	Years .exp
P1	Clinical psychologist	F	Private + hospital	20~
P2	Clinical psychologist / psychotherapist	F	Private + trauma center	+20
P3	Clinical psychologist	M	Private	+10
P4	Clinical psychologist (.trauma spec	F	Private + NGO	+7
SW1	Hospital social worker	F	Public hospital	30~
SW2	Hospital social worker	F	Public hospital	+5
W1	Welfare social worker	F	Municipal welfare	2.5~
W2	Welfare social worker	F	Municipal welfare	4
AI1	Clinical psychologist (LLM persona - (Gemini	M	Private clinic	10

AI2	Social Worker (LLM persona - Claude)	F	Municipal welfare	9
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4.2 Interviews Overview

The interviews with mental health professionals enabled us to identify key patterns in anxiety management, better understand patients' real needs, and define functional features that contribute to creating an effective, accessible, and personalized support system.

Table A1(Appendix A) consolidates fourteen key categories derived from the interview findings, each attributed to the relevant professionals. These categories span the full clinical spectrum: Common Symptoms of Anxiety (physical, emotional, and cognitive manifestations), Types of Anxiety (age-specific, trauma-related, and event-specific presentations), and Treatment Strategies (physiological, cognitive, and emotional interventions). The professionals emphasized that anxiety management requires sensory engagement through Grounding and Sensory Techniques (five-senses focus, present-moment awareness, and guided actions), delivered via Voice Guidance and Bot Features (warm tone, multilingual support, and authentic emotional connection). Critically, all interventions must be Personalized according to patient preferences and therapist guidance, with Therapist Involvement carefully calibrated—therapists should monitor remotely and provide guidance, but only within realistic availability constraints and after establishing personal familiarity with patients.

Across these fourteen categories, three consistent themes emerge:

- (1) the need for immediate, sensory-based intervention during an anxiety episode, supported by grounding techniques, haptic feedback, and emotionally intelligent voice guidance.
- (2) the importance of therapist oversight balanced against patient independence, requiring remote monitoring, risk detection, and graduated support rather than constant direct intervention.
- (3) strict boundaries around privacy and anonymity, particularly in community-facing features, prioritizing data minimization and controlled access over expansive data collection.

These themes directly informed the functional and non-functional requirements presented in Section 4.5.

4.3 Interviews Insights

Based on the professional interviews summarized in the table above, it is suggested that while anxiety manifests differently among patients, strong common patterns exist across different demographics. These clinical insights directly shape the required features and core functionality of the proposed anxiety-support system:

- **Common Symptoms:** Professionals confirmed that anxiety presents as a severe mix of physical (rapid heartbeat, shortness of breath, freezing), emotional/cognitive (intense fear, overthinking, loss of control), and behavioral symptoms (avoiding triggers and functional impairment).
- **Effective Coping Methods:** Immediate intervention is critical. Interviewees strongly advocated for controlled breathing, five-senses grounding techniques, and calming stimuli. This validates the app's need for accessible interventions like haptic (vibration) alerts and step-by-step calming audio guidance.
- **Continuous Monitoring and Tracking:** Simple tracking tools for attack frequency, timing, and severity help patients recognize their own anxiety patterns. Additionally, they allow the system to generate concise progress summaries for continuous remote therapist support.
- **Positive Reinforcement and Motivation:** To encourage app usage and prevent drop-off, the experience must be empathetic. Any digital bot must use a warm, calm tone and reassuring phrases (e.g., "I'm here with you") to provide emotional validation without feeling robotic.
- **Personalization:** Since anxiety ranges widely from general stress to severe trauma, the system must offer highly tailored tools. Users and therapists need the ability to customize coping mechanisms, preferred sensory tools, native language options, and the level of direct therapist involvement.

4.4 Challenge encountered during interview:

Understanding the Clinical Side:

- Clinical and Domain Understanding: One of the primary challenges at the beginning of the project was becoming familiar with the field of psychology. It is a complex domain that is not traditionally highly integrated with advanced digital development. Bridging this gap required a deep, thorough understanding of

clinical needs before we could accurately begin translating them into technological solutions.

- Research and Data Collection: During the research process, difficulties arose in recruiting professionals for interviews, which resulted in a relatively small sample of participants (five professionals). This limitation affected the diversity of the responses and made it more difficult to generalize the findings. Furthermore, we faced a significant challenge in consolidating the different interview responses into clear and structured categories due to the high variability in the interviewees' individual clinical perspectives and experiences.

Technology:

- On the engineering side, a central technical challenge was establishing hardware-to-software communication. Specifically, we had to determine how to reliably connect the wearable device to the application, how to transmit and receive continuous Bluetooth signals, and how to effectively parse and interpret the data contained within those signals. Determining how to accurately analyze these signals in real-time was crucial so the software could successfully detect anxiety and trigger the appropriate interventions.

4.5 Requirements

Building on the clinical insights from our expert interviews and competitive analysis, we have defined functional and non-functional requirements for the anxiety-support system. The requirements below represent the central features for Phase B development, organized around three key principles: enabling immediate intervention during anxiety episodes, supporting therapist oversight, and protecting user privacy. The complete set of detailed supporting requirements is documented in Appendices B and C.

4.5.1 Functional Requirements

Table 3. Functional requirements

No.		Requirement
1		The System shall support user registration and login
2		The System shall provide an initial settings process for patients
	2.1	The System shall allow patients to select a therapist from a list of available therapists during the initial settings process
	2.2	The System shall provide options during the initial settings for patients to select calming actions
3		The System shall enable patients to view and manage their personal progress dashboard
4		The System shall allow therapists to view reports on patient progress over specified time periods
	4.1	The System shall enable therapists to recommend customized relaxing actions
5		The System shall detect anxiety attacks based on physiological and/or behavioral indicators
6		The System shall provide calming actions during anxiety attacks
7		The System shall support documentation and monitoring
8		: The System shall provide motivational feedback and engagement incentives
9		The System shall allow users to join/view community posts related to anxiety coping and emotional support

4.5.2 Non Functional Requirements

Table 4. Non functional requirements

ID	Requirement	Metric	NFR Category
1	User Anonymity	100% of community posts shall be displayed without full names, phone numbers, email addresses, or other personally identifiable information (PII).	Security (Privacy)

2	Content and Interaction Security	100% of data transmitted between the client and server shall be encrypted using HTTPS/TLS.	Security
3	Unauthorized Access Prevention	100% of sensitive operations shall require user authentication and appropriate access permissions.	Security
4	Therapist Access to Non-Identifiable Data	100% of information accessible to therapists shall be anonymized and contain no personally identifiable information.	Security (Privacy)
5	Mental Health Safety	At least 95% of content identified as harmful or promoting unhealthy behavior shall be removed or hidden within 24 hours.	Safety
6	System Availability	Community services shall maintain at least 99% annual uptime, excluding scheduled maintenance periods.	Availability
7	System Performance	Community-related requests shall receive a response within 3 seconds under normal operating conditions.	Performance

4.6 Technology Stack

Hardware connectivity: The smart jewelry communicates with the mobile app over Bluetooth, transmitting sensor readings that indicate physiological signs of an anxiety attack so the app can respond in near real time.

Frontend technologies: The application is built using React Native, providing a single responsive interface for both patients and therapists across Android and iOS devices.

Data management: Firebase Firestore serves as the primary database, offering flexible document-based storage for patient profiles, anxiety episode logs, and community posts, along with real-time listeners that push updates directly to the app as soon as data changes.

Server-side logic: Firebase Cloud Functions replace a dedicated backend server, running small pieces of triggered code in response to Firestore events — for example,

automatically notifying a patient's assigned therapist when a new anxiety episode is documented, without needing to host or scale a separate server.

Authentication: Firebase Authentication manages secure sign-in for patients, therapists, and community managers, supporting role-based access enforced through Firestore Security Rules so that sensitive health data shared with a therapist stays scoped to that relationship.

File storage: Firebase Cloud Storage holds binary content such as recorded voice notes from calming sessions and any media shared within the community.

Push notifications: Firebase Cloud Messaging delivers alerts to the app, such as notifying a therapist when a patient's jewelry detects an anxiety attack, or notifying a patient when a community message goes unanswered.

4.7 Diagrams

The system's functionality is analysed from the perspective of four actors: the Patient, the Smart Jewelry, the Therapist, and the Community Manager. The Patient uses the system to view anxiety alerts triggered by the jewelry, use a range of calming actions during an episode, document and optionally share anxiety episodes with their therapist, set up a profile, and participate in the community by joining, posting, and reporting inappropriate content. The Smart Jewelry's role is limited to detecting anxiety indicators and transmitting sensor data to the app. The Therapist views patient reports, attack-frequency trends, and community insights, and can provide emergency support, which extends into viewing the relevant anxiety alert. The Community Manager moderates community content, with deleting a post modeled as an extension triggered by a patient's report rather than a routine action.

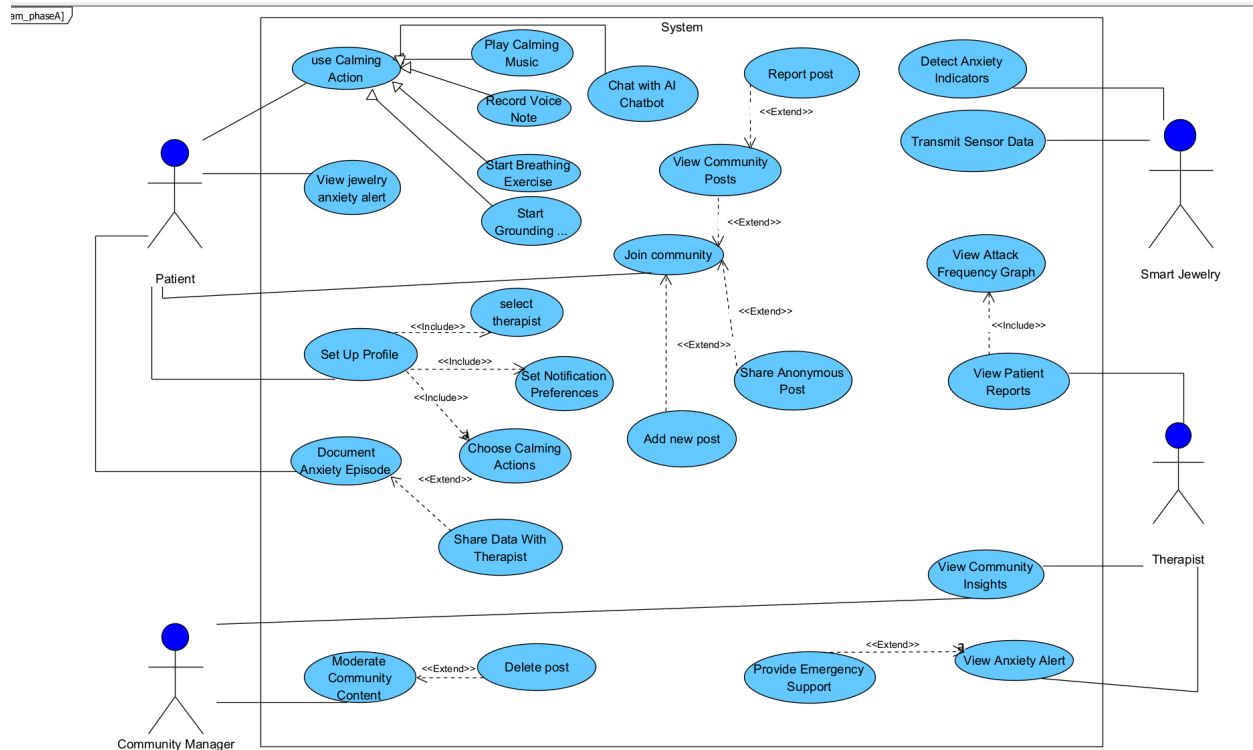


fig 2. Use case diagram for the anxiety-support jewelry application

Figure 3 shows how a patient browses community members, selects either a therapist or another patient, and sends a message. If the recipient is unavailable, control loops back so the patient can pick someone else or exit. The Community Patient and Community Therapist lanes show the receiving side: each views the incoming message and either replies, completing contact, or marks themselves unavailable, which notifies the sender to try again.

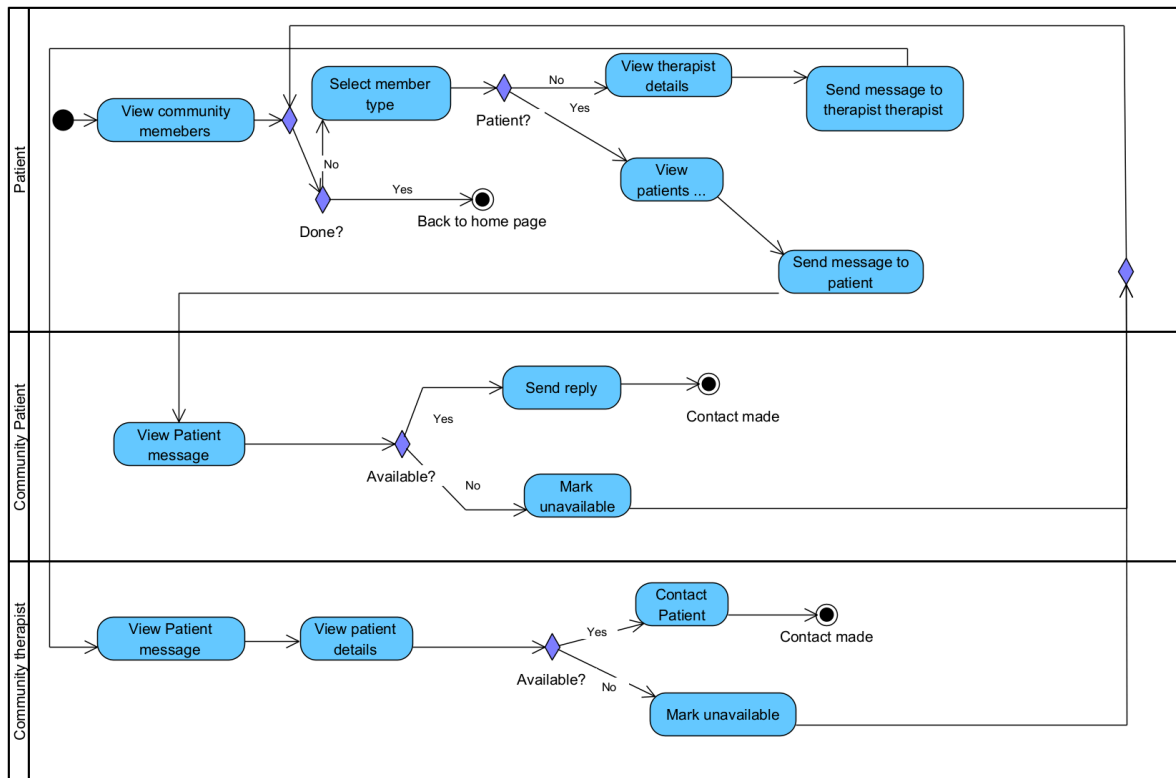


fig 3. Activity diagram - Patient and community

Figure 4 shows three participants: the jewelry, the patient, and the application. The jewelry detects an anxiety attack and notifies the app, which presents the patient with calming options. The patient selects one action — playing music, a breathing exercise, a podcast, or calling the therapist — and the app starts a calming session. The system then checks whether the anxiety has subsided: if not, the patient loops back to try another action; once resolved, the episode is documented.

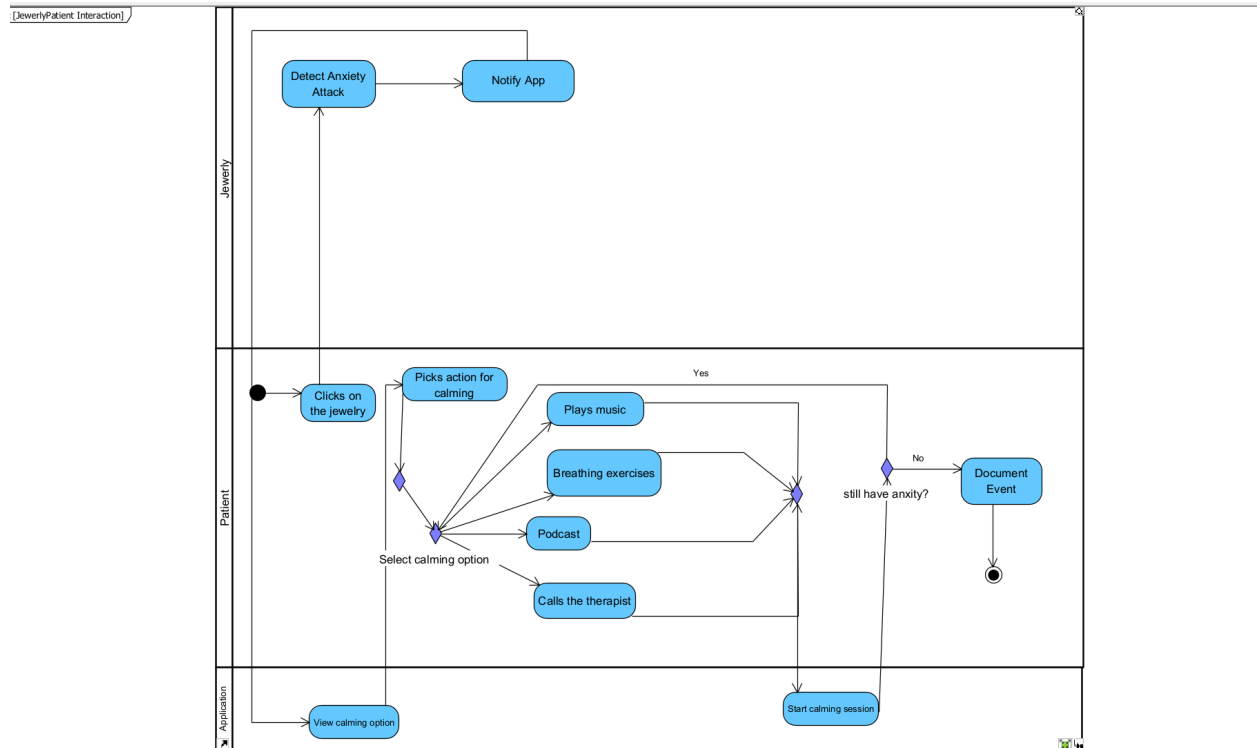


fig 4. Activity diagram - Patient and Jewelry

Figure 5 shows the split between the frontend and backend. On the frontend, the smart jewelry transmits data to the React Native app over Bluetooth. Both React Native and a React web app connect independently to Firebase using the same read, write, and live-update channels shown in the middle. On the backend, Firebase stores profiles, posts, and logs, and handles authentication for patient and therapist roles. Firestore writes trigger Cloud Functions, which send push notifications through Cloud Messaging, while Cloud Storage holds voice notes and other media uploaded from either client.

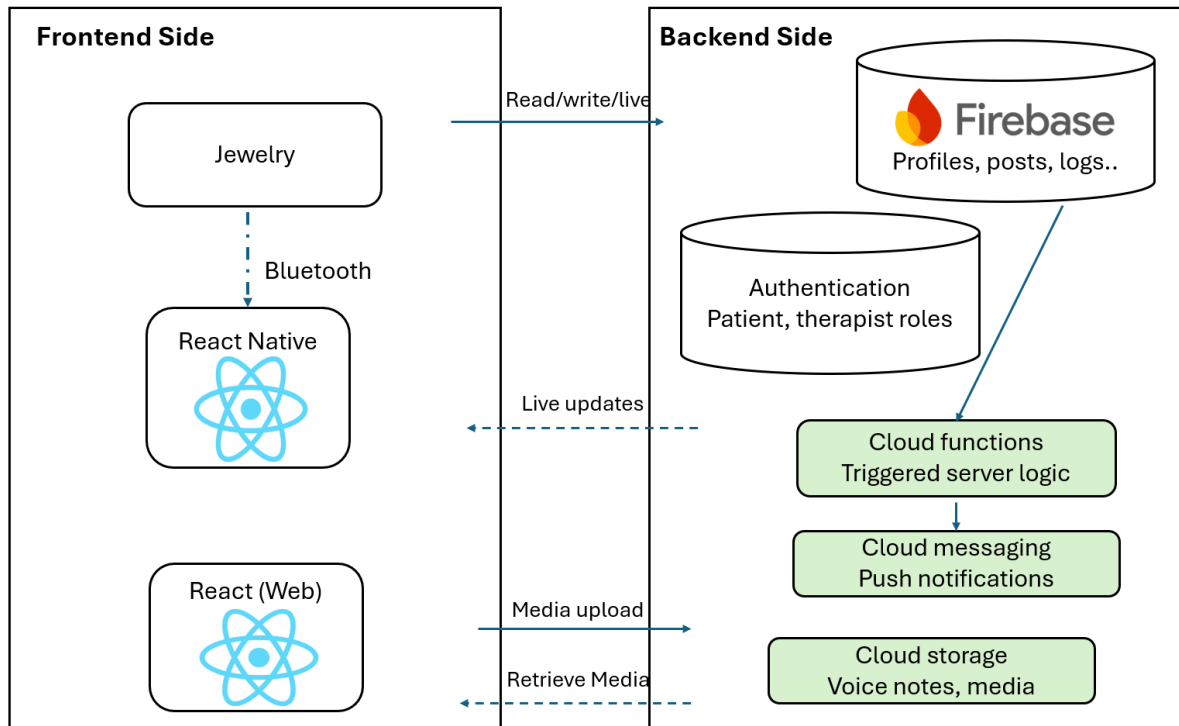


fig 5. System Architecture diagram

5. Tests and metrics and criteria

5.1 Success Metrics

Subjective Metrics (User Perception & Satisfaction):

- Badge and gamification engagement (users earning and redeeming points/badges)
- User satisfaction: 80% or higher satisfaction rating on System Usability Scale (SUS)
- Trust and confidence in the tool (measured through user feedback and clinical assessment)

Objective Metrics (Measurable System Performance):

- Chat interaction duration and frequency (therapist-patient communication engagement)
- Fidget/jewelry usage time (wearable engagement and system adoption)
- Amount of personal information entered into the application (indicator of user engagement and data utilization)
- User adoption rate: 70% adoption among patients and therapists

- Data security compliance: 99% compliance with data protection regulations and zero security breaches
- System scalability: Supporting at least 10,000 concurrent users without performance degradation
- Therapeutic outcomes: Measurable reduction in anxiety episode frequency and intensity through regular application use

5.2 Functional Tests

Functional testing will validate that the system meets the central requirements defined in Section 4.2. Key test areas include:

User Authentication and Setup

- User registration, login, and role-based access control
- Therapist selection and calming action preferences during initial setup

Patient Features

- Personal dashboard displays and updates anxiety data accurately
- Progress graphs and analytics work correctly
- Badge and incentive systems function as expected

Anxiety Detection and Intervention

- Wearable detection triggers alerts correctly
- Calming actions (music, breathing, chatbot, grounding) launch and respond appropriately

Documentation

- Note-taking, voice recording, and structured documentation work reliably
- Therapist access to patient documentation respects consent settings

Community Features

- User anonymity is maintained in community posts
- Commenting, reactions, and content reporting function correctly
- Content moderation and therapist oversight work as intended

System Integration

- Bluetooth communication between jewelry and app is stable
- Data synchronization and encryption between frontend and backend work reliably

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Appendix

Appendix A: Interview Insights by Category

Table A1: Categories and Key Insights with Professional Attributions

Feature	Key Insights with Professional Attributions
Common Symptoms of Anxiety	• Physical: Rapid heartbeat, shortness of breath (P1, SW1, SW3), and freezing response/functional impairment (SW4). • Emotional: Panic, emotional overload, loss of control (P1, SW4), intense fear, and emotional flooding (SW1, SW2, SW3). • Cognitive: Confused thoughts, difficulty concentrating (SW1), overthinking (SW2), and cognitive overload/loss of awareness (SW2, SW3).
Types of Anxiety	• Age-specific: Highly prevalent among adolescents and children (SW1, SW3) • Trauma & Crisis: Trauma-related, emergencies, family violence, and acute stress reactions (SW2, SW3, SW4) • Event-specific & Chronic: Fear of exams, emotional suppression, loneliness, and chronic episodes (P1, SW2).
Treatment Strategies	• Physiological/Sensory: Five senses grounding, deep breathing, water/ice, and muscle relaxation (P1, SW1, SW2, SW3). • Cognitive/Behavioral: Distraction, CBT techniques, Socratic questioning (SW1, SW2), and task-oriented behavioral activation (SW4). • Emotional/Supportive: Talking with trusted people (SW1), emotional validation, and supportive communication (SW4).

Grounding and Sensory Techniques	• Sensory Focus: Strong emphasis on the five senses—smell, touch, taste, hearing, sight (P1, SW3). • Present Moment: Redirecting attention away from thoughts and reconnecting to the present moment (SW1, SW2). • Action & Audio: Guided imagery, calming sounds/music, Quran (SW3), and regaining control through small actions/coping tasks (SW4).
Voice Guidance and Bot Features	• Tone & Empathy: Warm emotional voice using phrases like "I'm here with you" (P1) and "You are not alone" (SW2). • Language: Mother tongue support, correct pronunciation, and multilingual capabilities (SW1, SW3). • Human Connection: Human-like emotional connection is vital; a bot alone may not be enough (SW4).
Personalization	• Patient Preferences: The patient should choose preferred coping and sensory tools beforehand (P1) • Therapist Guidance: The therapist should define which techniques suit each specific patient (SW2). • Needs & Language: Personalized according to patient needs, language, music choices, and individualized emotional support (SW1, SW3, SW4).
Therapist Involvement	• Availability Limits: Depends on working hours and risk level (SW1); unsure about constant direct availability during emergencies (SW4) • Prior Familiarity: Therapists should know the patient personally before providing intervention (SW2) • Remote Tracking: Therapists can monitor remotely and receive updates (SW3), which is important during treatment stages (P1).

Risk Detection / Red Flags	• Risk Assessment: Based on severity/immediate help needs (P1), with red flags for self-harm, loneliness, and suicidal risk (SW1, SW2). • Trigger Sensitivity: Warning: asking about suicide may negatively influence younger users (SW3). • Silent Alerts: Red flag notifications should alert the therapist, not the patient (SW4).
Community Support	• Clinical Moderation: Helpful only if clinically supervised and moderated (P1, SW3) • Timing & Isolation: Best used outside of acute anxiety attacks (SW1) to help patients feel less alone, as shared experiences are beneficial (SW2, SW4).
Privacy Concerns	• Anonymity & Oversharing: Avoid oversharing, use nicknames, and avoid exposing private identities (SW2, SW3) • Data Minimization: Avoid requesting exact locations, highly personal info, or unnecessary details (SW1, SW4) • Controlled Access: Ensure controlled therapist involvement and limit dependency (P1).
Language and Accessibility	• Mother Tongue: Strongly recommended to use native language and culturally adapted content (SW1, SW3). • Emotional Resonance: Emotional language and reflective communication (P1), personalized to the patient's state (SW2), are more important than voice gender matching (SW4).
Monitoring and Progress Tracking	• Attack Data: Tracking attack timing, frequency, severity scales, and attack stages (P1, SW1). • Therapist Summaries: Providing continuous updates and summary information for therapy teams (SW2).

	• Patterns & Logs: Weekly tracking of symptom patterns, coping behaviors, and daily journaling (SW3, SW4).
Dependency Concerns	• Limiting Access: Warned against over-dependency; access should be limited unless high risk, as over-connection negatively affects treatment (P1, SW1, SW2) • Independence: The therapist should gradually encourage patient independence (SW3) • Supplemental Role: The app should support, but absolutely not replace, real human therapy (SW4).
Suggested App Features	• Immediate Interventions: Vibration alerts, sensory exercises (P1), and AI guidance for grounding (SW2). • Emergency Connections: Family contact, emergency communication (SW1), team-based therapist support, and human support integration (SW2, SW4). • Content Variety: Calming audio (SW1), music, podcasts, guided imagery, multilingual support (SW3), and structured coping tasks (SW4).

Appendix B : Additional functional requirements

No.	Requirement
1	The System shall support user registration and login
1.1	The System shall provide patient registration with personal settings
1.2	The System shall provide therapist registration with professional details
1.3	The System shall allow login for patients
1.4	The System shall allow login for therapists
2	The System shall provide an initial settings process for patients
2.1	The System shall allow patients to select a therapist from a list of available therapists during the initial settings process
2.2	The System shall provide options during the initial settings for patients to set notification preferences for anxiety episodes
2.2.1	The System shall allow patients to enable or disable vibration notifications
2.2.2	The System shall allow patients to enable or disable sound notifications
2.2.3	The System shall allow patients to enable or disable popup text notifications
2.3	The System shall provide options during the initial settings for patients to set notification preferences for the end of anxiety episodes
2.3.1	The System shall allow patients to enable or disable reminders to write documentation notes
2.4	The System shall provide options during the initial settings for patients to select calming actions
2.4.1	The System shall allow patients to optionally choose a relaxing song
2.4.2	The System shall allow patients to optionally choose breathing exercises
2.4.3	The System shall allow patients to optionally choose grounding exercises
2.4.4	The System shall allow patients to optionally choose between free-form or directed note documentation
2.5	The System shall allow patients to choose to share physical data of anxiety attacks, including intensity, duration, date, and time
2.6	The System shall provide an option for patients to choose to permanently share their written notes with their therapist
3	The System shall support therapist interactions and data access
3.1	The System shall allow therapists to add registered patients to their profile
3.2	The System shall allow therapists to view reports on patient progress over specified time periods
3.2.1	The System shall include a graph that reflects the frequency of attacks over time
3.2.2	The System shall include a graph of the intensity of attacks (duration)
3.2.3	The System shall provide an option to view the patient's documentation that the patient confirms to share

	3.3	The System shall enable therapists to customize calming texts for specific patient
	3.4	The System shall enable therapists to recommend customized relaxing actions
4		The System shall detect anxiety attacks based on physiological and/or behavioral indicators
	4.1	The System shall add an alert accompanied by text when the person experiences an anxiety attack
5		The System shall provide calming actions during anxiety attacks
	5.1	The System shall allow interaction with a relaxing chatbot
	5.2	The System shall allow playing calming music
	5.3	The System shall allow offering breathing exercises
	5.4	The System shall allow suggesting grounding exercises
	5.5	The System shall allow displaying soothing texts
6		The System shall support documentation and monitoring
	6.1	The System shall allow free note-taking
	6.2	The System shall allow structured note-taking
	6.3	The System shall allow a voice recording of the anxiety attack experience
	6.4	The System shall add an alert to document later when calmed down
7		The system shall provide a personal area for patients
	7.1	The System shall display personal settings
	7.2	The System shall allow updating of personal settings
	7.3	The System shall display graphs with the data of the attacks experienced
8		The system shall provide incentives when anxiety subsides
	8.1	The System shall display success or progress texts
	8.2	The System shall accumulate success coins based on indicators
	8.2.1	The system shall award coins based on whether the user filled out the note
	8.2.2	The system shall award coins based on whether the user performed sedative actions in the application during an anxiety attack
	8.2.3	The system shall award coins based on the duration of the attack
9		The System shall support storage
	9.1	The System shall securely store voice recordings
	9.2	The System shall securely store patient notes
10		The System shall support a community feature for users
	10.1	The System shall allow patients to join a community space with other users
	10.2	The System shall allow users to view community posts related to anxiety coping and emotional support

	10.3	The System shall allow users to share anonymized experiences of anxiety episodes
	10.4	The System shall ensure user anonymity when posting in the community
11		The System shall support user interaction within the community
	11.1	The System shall allow users to like or react to community posts
	11.2	The System shall allow users to comment on posts in a supportive manner
	11.3	The System shall allow users to report inappropriate content
12		The System shall provide moderation and safety mechanisms
	12.1	The System shall provide basic content moderation for community posts
	12.2	The System shall allow therapists or administrators to monitor community activity (if applicable)
	12.3	The System shall filter harmful or triggering content where possible
	12.4	The System shall enforce community guidelines to ensure a safe environment
13		The System shall provide motivational and supportive community features
	13.1	The System shall highlight positive and supportive posts
	13.2	The System shall display encouraging messages from the community
	13.3	The System shall allow sharing of coping strategies between users

Appendix C : Additional non functional requirements

ID	Requirement	Metric	NFR Category
1	Content Moderation	At least 95% of reported harmful content shall be reviewed by an automated moderation system or a moderator.	Safety
2	Report Handling	95% of user reports shall be reviewed and processed within 24 hours of submission.	Performance
3	Positive Communication Promotion	At least 90% of published comments remaining after moderation shall comply with community guidelines.	Safety
4	Personalization and Content Blocking	Blocking or hiding a user or content item shall be completed within 1 second.	Performance
5	Privacy Compliance	100% of privacy requirements defined for the project shall pass annual compliance audits.	Security (Privacy)